



Deep learning driven radiographic classification of primary bone tumors using attention augmented hybrid models

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ABSTRACT

Accurate classification of primary bone tumors is necessary for timely diagnosis and effective treatment planning, particularly given the complex radiographic heterogeneity exhibited by tumor subtypes. The present study introduces two novel deep learning models, including a Convolutional Neural Network Transformer (CNNT) hybrid and a Residual Network 50 (ResNet50) model, augmented by a Convolutional Block Attention Module (CBAM) to enhance feature discrimination and contextual understanding in radiographic images. The models are trained and validated on the Bone Tumor X-ray Radiograph Dataset (BTXRD) dataset of 3,746 labeled radiographs containing nine tumor subtypes. To counter the effects of noise and class imbalance, advanced preprocessing methods like Block Matching 3D Filtering (BM3D) and data balancing using the Synthetic Minority Over sampling Technique (SMOTE) are employed.

Extensive testing demonstrates that our approaches outperform current state of the art models, such as ResNet50, EfficientNet version B3 (EfficientNet-b3), You Only Look Once version 8 classification (YOLOv8s-cl), and Deep Supervision Network (DS-Net). Specifically, the ResNet50-CBAM architecture achieves an F1-score of 0.9759, an AUC-ROC score of 0.984, mean accuracy of CBAM 97.41% and a Cohen's Kappa score of 0.9718, outperforming existing benchmarks for binary tumor classification. The CNNT model also achieves competitive performance, reaching an F1-score of 0.9595 with an accuracy of 92.56%. Incorporating attention mechanisms and dataset guided preprocessing renders this framework appropriate for practical clinical settings. The findings of this research have significant implications for the healthcare sector by introducing a scalable, interpretable, and highly accurate Artificial Intelligence (AI) based diagnostic system that can support radiologists in timely diagnoses and decision making processes, ultimately contributing to better patient outcomes and alleviating the diagnostic burden in musculoskeletal oncology.

1. Introduction

Primary bone tumors, while rarely seen, are a significant health concern, particularly in people under the age of 20 years [1], since they are the third most frequent cause of cancer related death among this age group. Accurate and early diagnosis is part of the best therapeutic management and better patient prognoses. Radiographic imaging remains the cornerstone technique of initial evaluation for suspected bone tumors, providing invaluable information regarding the location, architectural features, and potential malignancy of the cancer. The highly variable radiographic presentations of bone tumors, however, pose complex diagnostic problems even to skilled radiologists [1].

Recent breakthroughs in Artificial Intelligence (AI), specifically those from deep learning methods, have radically changed the field of Computer Aided Diagnosis (CAD) in medical imaging. These AI based

methods have exhibited unprecedented performance in various tasks, such as image segmentation, detection, classification, and enhancement in various medical applications. In the field of musculoskeletal oncology, studies have proven the effectiveness of deep learning models in the detection and evaluation of bone tumors using radiographic images, a significant advancement in the field of diagnostic radiology. For instance, a multitask deep learning model by von Schacky et al. achieved an accuracy of 80.2% in classifying primary bone tumors as malignant or benign, outperforming radiology residents but matching those of fellowship trained radiologists [2].

The clinic's integration of AI diagnostic tools is hindered, among other things, by a lack of large, well annotated, publicly available radiographic datasets for bone tumors. The rarity of bone tumors such

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as Osteosarcoma results in small case numbers, which are predominantly centered in large teaching hospitals. Patient privacy issues also surround medical images, making using permissions at a large scale difficult. Further, image annotation is time and labor intensive, with experienced physicians taking a long time. This absence of robust datasets limits AI model development, validation, and generalization, restricting clinical use [2]. Yu He et al. developed a 2899 X-ray image database at five centers, 2158 non malignant and 741 malignant. Images did not have significant clinical information such as gender, age, and geographic location. Chonnam National University Hospital (CNUH) dataset includes 1195 tumor images and 381 standard images but is limited to knee bone tumors, so its applications are limited. Others, such as Dan Yu, Chan Woo Park, and Sarah Consalvo, have the same drawbacks, such as limited sample size, standard bone images, and demographic information [3].

An effort is made to address these problems [2]. The Bone Tumor X-ray Radiograph Dataset (BTXRD) was created to support AI algorithm development in diagnosing primary bone tumors. The dataset includes 3746 bone images, tumors, and typical cases, along with clinical data and annotations [1]. The Bone Scan 80K Dataset (BS-80K) dataset has many bone scan images for computer aided diagnosis research. However, overall dataset availability is limited, indicating the need to improve data collection further and share to develop AI in bone tumor diagnoses [3].

The use of artificial intelligence (AI) for bone tumor classification has a long history of development, from the initial work of Lodwick et al. [4] in 1980 with early computer models for tumor evaluation to Reinus et al. [5] in 1994, who used neural network based models which translated imaging features of 709 bone lesions into structured databases, achieving diagnostic accuracy of 85% in differentiating benign from malignant conditions. Next, Do et al. [6] presented a Bayesian inference model in 2017, achieving 62% accuracy in predicting precise diagnoses and 80% in differential diagnosis from 710 radiographs Do et al. In addition to classification, AI has become highly prevalent in tumor boundary segmentation. For example, Huang et al. [7] designed a Multiple Supervised Fully Convolutional Network (MSFCN) with a Dice Similarity Coefficient (DSC) of 87.8% and F1-score of 0.908 superior to Fully Convolutional Network (FCN), U-shaped Convolutional Network (U-Net), and Hypercolumn Edge Detection (HED) models in segmentation Huang et al. [7]. Subsequent model MSRN further enhanced segmentation metrics by 8.78% DSC over state of the art baselines Do et al. More recent work has also ventured into predictive modeling for postoperative tumor recurrence: He et al. [8] applied CNNs to Magnetic Resonance Imaging (MRI) data from 56 giant cell tumor patients, with CNN accuracies of 75.5% and CNN + regression accuracies of 78.6%, exceeding radiologists' 64.3% accuracy, with sensitivity gains of more than 20% He et al. [9]. These examples collectively demonstrate AI's advancing sophistication and potential for bone tumor diagnosis, localization, and prognosis.

Despite these developments, several significant limitations are still present in the existing literature. Several older and newer models are plagued by poor dataset diversity, absence of demographic metadata, and incomplete subtype annotations, which restrict their generalization capacity [10]. Researchers such as von Schacky et al. and Yao et al. utilize rich information datasets; however, they do not contain a complete multiclass taxonomy or consistent high resolution annotations and are frequently restricted to particular anatomical areas (e.g., solely knee specific) [1,2]. Moreover, traditional architectures such as Residual Network 50 (ResNet50) or Inception V3, although suitable for general use, fail to distinguish visually comparable tumor subtypes due to their inability to grasp the general context and the intricate local features [10,11]. Lastly, few studies have used advanced denoising methods or addressed the extreme class imbalance commonly encountered in bone tumor datasets [12].

Our work proposes a dual model solution to address these gaps: a CNN Transformer hybrid (CNNT) and a ResNet50 model with Convolutional Block Attention Module (CBAM). Our models are trained on the

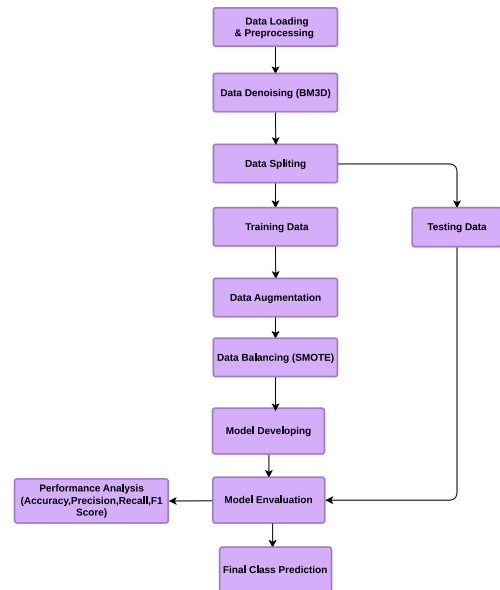


Fig. 1. The proposed framework.

BTXRD dataset, a strong, multicenter dataset of 3746 radiographs of nine bone tumor subtypes and annotated demographic, anatomical, and imaging metadata [1]. Our solution also incorporates BM3D filtering a state of the art denoising method tested on Computed Tomography (CT), MRI, and histopathology imaging modalities and SMOTE over-sampling to handle class imbalance efficiently. The CBAM enhanced ResNet50 detects critical spatial and channel wise dependencies, while the CNNT's transformer module is particularly well tuned in modeling long range contextual relationships. These improvements achieve significant gains over recent benchmarks (e.g., You Only Look Once version 8 classification (YOLOv8s-cls), EfficientNet, Deep Supervision Network (DS-Net), with an F1-score of 0.9759 and Cohen's Kappa of 0.9718 comfortably outperforming records by a wide margin [1,3,12]. The following are the key contributions of our proposed work.

1. Proposing a two path framework combining CNN, CBAM, and Transformer for efficient local and global feature modeling.
2. Utilizing BM3D denoising and SMOTE balancing for enhancing image quality and addressing class imbalance.
3. Use of the comprehensive BTXRD dataset with rich clinical metadata for better generalization.
4. Achieving superior performance in multiclass classification metrics (AUC-ROC, precision, F1-score, specificity).
5. Evaluating against ResNet50, EfficientNet, YOLOv8s-cls, and DS-Net yields state of the art outcomes.
6. Achieving effective training with lower complexity and quicker convergence, appropriate for clinical use in real world environments.

The rest of the paper is organized as follows. Section 2 covers data preparation and preprocessing, Section 3 covers the proposed methodology, Section 4 presents the results and discussion, and Section 5 concludes the paper with future research directions

2. Data preparation and preprocessing

A stepwise description of the methodology adopted by this research from data acquisition to model assessment is illustrated in Fig. 1.

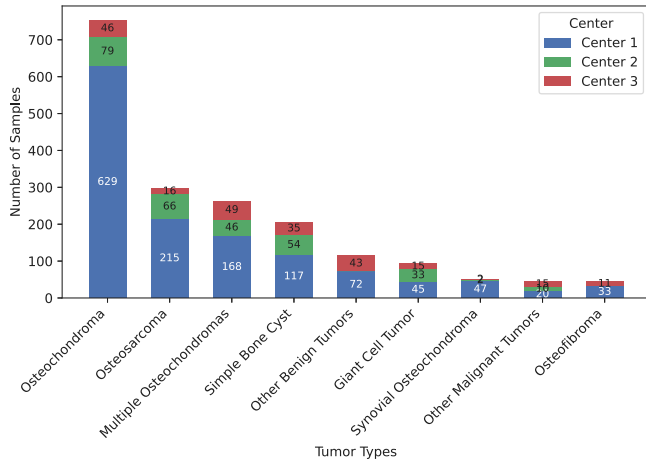


Fig. 2. Distribution of bone tumor subtypes in the BTXRD dataset.

It outlines the step by step pipeline, which includes image preprocessing, feature extraction, attention enhancement, classification, and performance assessment.

Fig. 1 presents the overall methodology workflow concerning the proposed study. The process starts with acquiring the BTXRD dataset and then preprocessing the data, including denoising using BM3D, augmentation, and balancing using SMOTE. The preprocessed images are then passed through two hybrid deep learning models: CNNT and ResNet50-CBAM. These models perform feature extraction and classification. The models are evaluated using accuracy, F1-score, AUC-ROC, MCC, and confusion matrix evaluation. The following subsections provide details of the dataset and the pre processing steps before the data can be fed into the proposed deep learning models.

2.1. Dataset

In this work, we used the BTXRD dataset, an open dataset for the classification, localization, and segmentation of primary bone tumors from radiographs [13]. The dataset consists of 3746 X-ray images, including 1879 standard bone images, 1525 benign bone tumor images, and 342 malignant bone tumor images. Each image also annotates patient demographics (age, gender), anatomical region, and imaging parameters (shooting angle). Besides this, bone tumor images also contain benign/malignant status, subtype classifications, and anatomical site annotations to maintain complete categorization. The data were gathered from three extensive data collections: (1) three Chinese medical centers, namely First Affiliated Hospital of Guangxi Medical University, Affiliated Hospital of Youjiang Medical College for Nationalities, and Baise People's Hospital; (2) Radiopaedia.org, a popular web radiology portal; and (3) MedPix, an open medical image database. For data consistency assurance, primary bone tumor samples involving the head, neck, chest, and abdominal regions were rejected due to the low availability and occurrence of imaging artifacts. Therefore, the dataset is concentrated towards X-ray images of bone tumors of the upper limb (hands, ulna, radius, humerus, wrist joints, elbow joints, and shoulder joints), lower limb (feet, tibia, fibula, femur, ankle joints, and knee joints), and pelvis (hips, hip joints). The annotation and labeling work was carried out by domain professionals, where one bone tumor specialist with 10 years of diagnosis experience carried out the initial segmentation using the Label me [14] annotation tool and tumor marking using bounding boxes. Then, the other expert with 20 years of experience reviewed it for accuracy verification. In addition, 50 of the images randomly selected were resegmented after a month to check the reliability of the annotations. All dataset collection, and

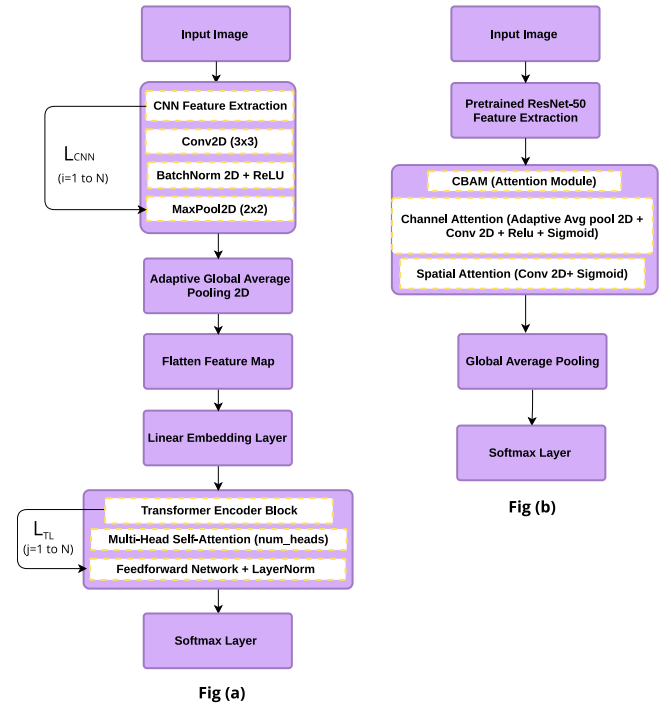


Fig. 3. (a) CNNT: Convolutional Neural Network Transformer model integrating local feature extraction through CNN layers and global context modeling via multi head self attention. (b) ResNet50-CBAM: A ResNet50-based backbone augmented for features extractions with a integrated Convolutional Block Attention Module (CBAM) to enhance spatial and channel wise attention for discriminative feature representation.

labeling procedures were conducted by the dataset authors, as cited in [13], and were not performed in this study. Our research focuses on primary bone tumor classification into nine different classes of tumors: Giant Cell Tumor, Multiple Osteochondromas, Osteochondroma, Osteofibroma, Osteosarcoma, Other Benign Tumors, Other Malignant Tumors, Simple Bone Cyst, Synovial Osteochondroma, presented in Fig. 2. Among these, Osteochondroma, Osteosarcoma, Multiple Osteochondromas, and simple bone cyst are the most common subtypes within the data.

The BTXRD dataset was meticulously curated through a stringent, multi stage process aimed at minimizing potential sources of bias and enhancing data integrity. High quality radiographic data were selectively acquired from diverse clinical centers, with particular scrutiny applied to data from Centers 2 and 3, which were identified as potential sources of confounding variability. Comprehensive preprocessing protocols including data cleaning, filtering, and validation were applied to exclude incomplete, low resolution, or redundant images, ensuring the inclusion of only diagnostically relevant samples. Clinical and demographic metadata were standardized into a uniform schema to facilitate cross source consistency and interoperability. Expert radiologists conducted manual annotations of bone tumor radiographs through a multi tiered review framework, generating high precision segmentation masks and bounding boxes. This rigorous annotation strategy was designed to minimize inter observer variability and improve the reliability of ground truth labels. To further strengthen the dataset's utility for deep learning applications, segmentation an essential task for isolating clinically significant anatomical structures was implemented using YOLOv8s-seg [15]. This state of the art, real time instance segmentation model combines object detection with pixel level mask

generation in a single stage architecture. Its ability to simultaneously localize bounding boxes and predict segmentation masks contributes to both computational efficiency and precise delineation of pathological regions. Collectively, these measures establish BTXRD as a robust and high fidelity benchmark for the development and evaluation of automated bone tumor diagnostic models [16].

The BTXRD dataset consists of nine tumor subtypes with varying class frequencies. The subtype distribution in the dataset is presented in Fig. 2.

Fig. 2 shows the class wise distribution of bone tumor subtypes in the BTXRD dataset. It can be observed from the figure that there exists class imbalance e.g., *Osteochondroma* and *Simple Bone Cysts* contain much larger sample counts than those occurring infrequently, like *Osteofibroma*. This motivated our application of SMOTE to alleviate skewness and enhance model learning over minority classes.

2.2. Data denoising

Medical image datasets are usually noisy due to variations in acquisition conditions, sensor defects, and other external conditions. Noise could degrade the performance of deep learning models, and thus strong denoising methods are required for preprocessing. Our research uses the Block Matching and 3D Filtering (BM3D) algorithm [17] to denoise the bone tumor images dataset before dividing it into training and testing subsets. BM3D is a cutting edge denoising method that builds similar image patches, converts these into a higher dimensional space, performs shrinkage filtering operations, and applies an inverse transformation to construct a denoised image. The algorithm efficiently preserves structural details while eliminating noise, making it very effective for medical image processing.

Recent studies have indicated the advantages of BM3D in biomedical imaging operations. For example, Zhang et al. [18] successfully applied BM3D to improve the quality of low dose CT scans, contributing to improved diagnostic accuracy. Similarly, Liu et al. [19] applied BM3D and deep learning architectures to denoising histopathological images and achieved enhanced feature extraction and classification performance. Wang et al. [20] applied BM3D based preprocessing methods to improve retinal fundus images, which contributed to enhanced segmentation performance for ophthalmological disease detection in one study. In addition, hybrid methods integrating BM3D and deep learning models have been researched, as illustrated in the research work of Chen et al. [21], where BM3D was used to denoise brain magnetic resonance scans prior to feature extraction and improve classification reliability.

In our approach, each image is preprocessed with BM3D before the dataset is divided into train and test splits. The algorithm is run independently for the Red Green Blue color channels (RGB) with predetermined noise parameters as defined in Algorithm 1. Denoised images are stored for subsequent model training, thus avoiding noise distortions that affect classification performance. The application of BM3D in our preprocessing procedure dramatically improves the quality of the tumor images dataset, thus improving the reliability and precision of tumor classification. Adaptive BM3D strategies and deep learning based denoising architectures can be explored in future research to further improve image quality and maximize classification accuracy.

We utilized a denoising pipeline using the BM3D approach to provide high quality input to the model training. The step by step procedure of denoising across the dataset over the RGB channels using BM3D is provided in Algorithm 1.

Algorithm 1 BM3D Based Image Denoising

Require: Input directory I_{input} , Output directory I_{output} , Noise parameters $(\sigma_r, \sigma_g, \sigma_b)$

Ensure: Denoised images are saved in I_{output}

- 1: **Initialize** $\sigma_r = 10, \sigma_g = 12, \sigma_b = 14$
- 2: Create I_{output} if not exists
- 3: Set counter $C = 0$
- 4: **for** each image $F \in I_{input}$ **do**
- 5: **if** F is valid image format **then**
- 6: Load I from F ; Skip if error
- 7: $(R, G, B) \leftarrow \text{split}(I)$
- 8: $(R, G, B) \leftarrow \text{BM3D}(R, \sigma_r), \text{BM3D}(G, \sigma_g), \text{BM3D}(B, \sigma_b)$
- 9: Save merged (R, G, B) as $I_{denoise} \in I_{output}$
- 10: $C \leftarrow C + 1$; Print success message
- 11: **end if**
- 12: **end for**
- 13: Print "Total images processed: C "

Algorithm 1 illustrates the denoising of the radiographic images through Block Matching and 3D Filtering (BM3D). Denoising is performed on every image in the dataset at certain noise levels. The resulting outputs are combined and saved as denoised images and used for training and testing, thus increasing feature interpretability and reducing classification artifacts.

2.3. Data augmentation

To enhance the robustness and generalization of our deep learning model for tumor classification, we employ a variety of data augmentation and preprocessing methods. Data augmentation is essential for increasing diversity in the dataset, reducing overfitting, and enhancing model performance.

The transformation pipeline starts with resizing the images to a target resolution, followed by the addition of random horizontal flipping to mimic natural variability in various bone tumor images. Random rotation within an interval of 15 degrees is also added, along with color jittering that involves controlled brightness, contrast, saturation, and hue adjustments. For added robustness, random perspective distortion is added, which comes in the form of a distortion scale of 0.2 with a probability of 0.5, thereby facilitating the generalization of the model over minor spatial deformations. Gaussian blur with a kernel size of 3 is also added to reduce variations in the sharpness of images. The images are finally converted to tensors and normalized to a mean of 0.5 and standard deviation of 0.5 to stabilize training.

2.4. Data balancing

Class imbalance is a significant problem in medical image analysis that can skew predictive models toward poor performance in minority classes. This problem is especially applicable in datasets for uncommon congenital abnormalities. To mitigate this problem, we use the Synthetic Minority Over sampling Technique (SMOTE) [22], a typical data augmentation technique that creates synthetic instances by interpolating between existing minority class instances. In our case, we use SMOTE to balance class distribution among various bone tumor images so that the model has been trained on a more representative set. SMOTE mitigates bias in imbalanced datasets by creating synthetic instances for minority classes instead of replicating instances, avoiding overfitting and improving generalization. The algorithm chooses random minority class instances and creates synthetic instances along the line segments between the k nearest neighbors in the feature space, thus increasing the representation of underrepresented classes without preserving the original data structure.

Recent studies have demonstrated the efficacy of SMOTE in enhancing classification performance for imbalanced biomedical data. For instance, Zhao et al. [23] applied SMOTE to dermatological image data, improving classification accuracy for rare skin diseases. Similarly, Lee et al. [24] applied SMOTE based augmentation techniques to attain balance in pediatric craniofacial anomaly data, reflecting significant improvement in the stability of deep models. Additionally, SMOTE has been used in deep learning models in biomedical issues, including diagnosing rare diseases [25] and detecting congenital abnormalities [26], demonstrating its effectiveness in compensating for imbalanced distributions. For instance, [27] improved performance in deep models for detecting rare pathologies when SMOTE was applied in the preprocessing phase. Similarly, [28] demonstrated the effectiveness of SMOTE in addressing high dimensional biomedical data, which is the scenario for our application. While SMOTE involves some computational expense, its effectiveness in correcting biased distributions makes it worthwhile, as demonstrated by its widespread application in recent literature [28,29].

In our implementation, we use SMOTE on the indexed dataset, over-sampling minority classes to get an approximately uniform distribution before training. This ensures that the generalization of the classification model is improved, preventing the bias caused by imbalanced class distributions. By incorporating SMOTE in the preprocessing step, our method ensures that the proposed bone tumor mages dataset can be effectively used for medical diagnosis without being compromised by biased class distributions.

The basic architectural structure of the two models CNNT and ResNet50-CBAM is presented in Fig. 3.

3. The proposed models

First, we split the dataset into training and test sets. This allowed us to test the deep learning model without bias accurately. We took an 80:20 ratio, with 80% used for training and 20% for testing. We used the SMOTE on the training set alone to address the class imbalance issue, ensuring that no synthetic samples or augmented images leaked into the test set.

3.1. Convolutional Neural Networks Transformer (CNNT) model

The training process is designed systematically, starting with the initialization of models, where a CNN module generates spatial features from a sequence of convolutional layers followed by batch normalization, Rectified Linear Unit (ReLU) activation, and max pooling. The obtained feature maps are passed through an adaptive average pooling layer to preserve a fixed size representation, which is then transformed into a D-dimensional embedding space. The embeddings are passed on as input to the Transformer encoder, consisting of several layers with the multi head self attention mechanisms integrated to effectively capture the long range dependencies and contextual relationships in the images. The final classification is achieved through a fully connected layer with softmax activation for producing probabilistic outputs against the categories of various bone tumor images. The proposed CNNT model architecture is shown in the algorithm 2. The proposed model comprises several key elements: convolutional feature extraction, adaptive pooling, linear embedding, and transformer based attention modeling, followed by classification layers as shown in Fig. 3.

To increase the efficiency of the learning process, we utilize the Adam optimizer with a fixed learning rate, and the loss function is defined using categorical cross entropy. During the training process, the dataset is split into mini batches. For each batch of individuals, the model makes predictions, calculates the loss, and updates the

network parameters with backpropagation and gradient descent. Performance metrics such as Accuracy, Precision, Recall, and F1-score, as well as AUC-ROC, are calculated at the end of each epoch to monitor training and test performance. The model is subsequently exposed to an evaluation process, wherein it is switched to inference mode and executed on unseen test samples. The predicted outputs are compared with the ground truth labels, resulting in the calculation of the key performance metrics. This organized training and evaluation process helps to develop a robust and generalizable classification model capable of discriminating between various bone tumor subtypes such as Giant Cell Tumor, multiple Osteochondromas, Osteochondroma, Osteofibroma, Osteosarcoma, Other Benign Tumors, Other Malignant Tumors, Simple Bone Cyst, Synovial Osteochondroma

The model processes an input bone tumor image $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and number of channels, respectively. Feature representation is accomplished by a module composed of multiple convolutional layers, batch normalization, and non linear activation functions using a Convolutional Neural Network (CNN) based module. The feature representation is done as follows:

$$F_{\text{cnn}} = \text{ReLU}(\text{BatchNorm}(\text{Conv2D}(I, W_{\text{cnn}}))) \quad (1)$$

where W_{CNN} are learnable parameters of the convolutional kernels. This operation extracts local spatial features while retaining significant bone image structure patterns. A MaxPooling operation $\text{MaxPool2D}(2 \times 2)$ is then applied, subsampling spatial dimensions to pick dominant activations and thus improving computation efficiency and retaining significant structural information. The network learns hierarchical features from low level edge information to high level shape representations through several convolutional layers. The features are fed into the transformer based attention mechanism, further refining feature dependencies based on long range spatial relationships.

The feature extraction, an Adaptive Global Average Pooling (GAP) layer aggregates spatial information into a compact global descriptor. Mathematically, this operation is expressed as:

$$F_{\text{gap}} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F_{\text{CNN}}(i, j) \quad (2)$$

where H and W denote the spatial dimensions of the feature map, and $F_{\text{CNN}}(i, j)$ represents the activation values at position (i, j) . Unlike standard fully connected layers, GAP enables scale invariant learning of representations by compacting spatial features without losing essential patterns.

The feature map produced is flattened to a one dimensional vector to facilitate compatibility with the transformer based encoding module. The transformer encoder comprises multi head self-attention and feedforward layers that capture long distance dependencies between feature dimensions. Self attention mechanisms learn to scale feature interactions to facilitate strong contextual representation learning dynamically. Mathematically, the self attention mechanism calculates:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

where Q , K , and V are the query, key, and value matrices derived from the feature vector, and d_k is the scaling factor.

To capture long range dependencies bone tumor, we add a Transformer Encoder Block to process the extracted feature vector. The feature vector is first embedded through a linear embedding transformation to project it to a higher dimensional space

$$F_{\text{embed}} = W_e F_{\text{gap}} + b_e \quad (4)$$

where W_e and b_e are learnable parameters. This conversion enables compatibility with the transformer module by placing feature representations.

The transformer encoder utilizes Multi Head Self Attention (MHSA) to learn intricate inter dependencies between different feature elements. The self attention mechanism operates as follows

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

where Q, K, V are the query, key, and value matrices derived from F_{embed} , and d_k represents the dimensionality of the feature space. The self attention mechanism allows the model to assign different weights of importance to different spatial areas in the input feature map and thus capture long range dependencies effectively.

After MHSA, the transformer encoder employs a Feedforward Network (FFN) consisting of two fully connected layers with non linear activation

$$F_{\text{trans}} = \text{ReLU}(W_1 F_{\text{embed}} + b_1)W_2 + b_2 \quad (6)$$

This design improves the representation of learned features by focusing on the key structural patterns and eliminating redundant noise. Layer Normalization is applied between the feedforward and attention layers to allow stable training and better convergence.

By integrating CNN based local feature extraction and transformer based global modeling, the proposed framework is competent in accurately representing spatial and contextual information. This hybrid approach advances the classification of various bone tumor subtypes such as Giant Cell Tumor, multiple Osteochondromas, Osteochondroma, Osteofibroma, Osteosarcoma, Other Benign Tumors, Other Malignant Tumors, Simple Bone Cyst, Synovial Osteochondroma, which ensures accurate medical diagnosis.

The final representation from the transformer encoder is passed through a fully connected layer followed by a softmax activation function for multi class classification

$$\hat{y} = \text{softmax}(WF_{\text{trans}} + b) \quad (7)$$

where W and b are parameters to be learned, and $\hat{y}$ is the class probabilities for different bone type tumor. To improve the model's performance, categorical cross entropy loss is used

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (8)$$

where N is the number of classes which is 5, y_i is the ground truth label, and $\hat{y}_i$ is the predicted probability for each class. This specific loss function encourages high probabilities to be assigned to the correct class and penalizes incorrect predictions. Gradient based optimization strategies are utilized to train the model, allowing effective learning of precise nine type of tumor classification.

The training process uses the Adam optimizer with a starting learning rate of 10^{-4} , a batch size of 32, and early stopping based on test loss to avoid overfitting. The model is trained for 60 epochs, with weight updates performed through backpropagation.

$$W \leftarrow W - \eta \frac{\partial L}{\partial W} \quad (9)$$

where η represents the learning rate, and $\frac{\partial L}{\partial W}$ is the gradient of the loss function with respect to the model parameters.

Regularization techniques such as dropout and weight decay are applied to enhance generalization and mitigate overfitting. Dropout randomly deactivates a fraction of neurons during training, preventing

coadaptation of features. Weight decay (L2 regularization) discourages large weight values by adding a penalty term to the loss function.

The model's performance is continuously monitored using validation accuracy and loss. If validation loss stagnates or increases over successive epochs, early stopping halts training, preserving the best performing weights. This strategy ensures the model remains efficient while avoiding unnecessary computations, leading to the optimal classification of tumor.

The proposed CNNT model is learned by a properly designed pipeline that integrates global and local feature modeling. The extensive training process is shown in Algorithm 2.

Algorithm 2 CNN + Transformer Model Training for Bone Tumor Classification

Require: Image dataset D with labels $\mathcal{Y}$, batch size B , learning rate η , number of epochs E

Require: Number of CNN layers L_{CNN} , CNN neurons N_{CNN} , Transformer layers L_{trans} , number of heads H , hidden dimension D , number of classes N_c

```

1: Model Initialization:
2: Define CNN layers with  $L_{CNN}$  convolutional blocks, each with Conv2D, BatchNorm, ReLU, and MaxPool
3: Apply Adaptive Average Pooling to output size (8, 8)
4: Define embedding layer to map CNN output to  $D$ -dimensional space
5: Construct Transformer Encoder with  $L_{trans}$  layers and  $H$  attention heads
6: Define classifier: Fully connected layer with Softmax activation
7: Initialize HybridCNNTransformer model  $\mathcal{M}$ 
8: Define Loss Function and Optimizer:
9: Set loss function as categorical cross entropy
10: Use Adam optimizer with learning rate  $\eta$ 
11: Model Training:
12: for  $epoch = 1$  to  $E$  do
13:   Set model  $\mathcal{M}$  to training mode
14:   Initialize training loss  $\mathcal{L}_{train} = 0$ , correct predictions  $C_{train} = 0$ 
15:   Start training timer
16:   for each batch  $(X_b, Y_b)$  in  $D_{train}$  do
17:     Move  $(X_b, Y_b)$  to device
18:     Zero gradients of optimizer
19:     Compute model output:  $\hat{Y}_b = \mathcal{M}(X_b)$ 
20:     Compute loss:  $\mathcal{L} = \text{CrossEntropyLoss}(\hat{Y}_b, Y_b)$ 
21:     Perform backpropagation and update weights
22:     Accumulate training loss and correct predictions
23:   end for
24:   Stop training timer
25:   Compute training accuracy:  $\text{Acc}_{train} = \frac{C_{train}}{|D_{train}|} \times 100$ 
26:   Model Evaluation:
27:   Set model  $\mathcal{M}$  to evaluation mode
28:   Initialize test loss  $\mathcal{L}_{test} = 0$ , correct predictions  $C_{test} = 0$ 
29:   Start evaluation timer
30:   for each batch  $(X_b, Y_b)$  in  $D_{test}$  do
31:     Move  $(X_b, Y_b)$  to device
32:     Compute model output:  $\hat{Y}_b = \mathcal{M}(X_b)$ 
33:     Compute loss:  $\mathcal{L} = \text{CrossEntropyLoss}(\hat{Y}_b, Y_b)$ 
34:     Accumulate test loss and correct predictions
35:   end for
36:   Stop evaluation timer
37:   Compute test accuracy:  $\text{Acc}_{test} = \frac{C_{test}}{|D_{test}|} \times 100$ 
38:   Print epoch results: Training Loss, Training Accuracy, Training Time, Test Loss, Test Accuracy, Test Time
39: end for
40: Compute accuracy, AUC-ROC, precision, Recall, and F1-score
41: Other statistical Parameters

```

Algorithm 2 describes the training process of the CNNT model. It involves model initialization with CNN and Transformer layers, then mini batch training using categorical cross entropy loss. Forward pass, loss calculation, backpropagation, and weight updates are performed in each epoch. Performance is monitored using accuracy, AUC-ROC, F1-score, and other important metrics. Evaluation is performed after every epoch using unseen test data.

3.2. ResNet50 and Convolutional Block Attention Module (CBAM) model

The proposed ResNet50 and Convolutional Block Attention Module (CBAM) model is designed to enhance the classification performance of various bone tumor images by integrating a pre trained ResNet50 backbone and the CBAM. The integration effectively enhances feature extraction by channel and spatial attention mechanisms. The training process follows a systematic framework, as outlined in algorithm 3, including model initialization, loss function selection, optimization steps, training steps, and then testing. The model is initialized by loading ResNet50 with pre trained weights except for its last fully connected layer. The CBAM module is then integrated to highlight the most discriminative features in the images by the adaptive weighting of spatial and channel wise feature representations. This mechanism enhances the ability of the model to focus on discriminative regions of the bone images, which is essential in distinguishing between various bone tumor subtypes such as Giant Cell Tumor, multiple Osteochondromas, Osteochondroma, Osteofibroma, Osteosarcoma, Other Benign Tumors, Other Malignant Tumors, Simple Bone Cyst, Synovial Osteochondroma. A fully connected classifier is utilized to improve classification effectiveness, including batch normalization, ReLU activation, dropout, and softmax activation, thus ensuring effective and stable learning.

To improve the training process, categorical cross entropy loss with label smoothing is utilized, which reduces the impact of noisy labels and stops the model from showing overconfidence in its predictions. The AdamW optimizer is used due to its ability to optimize complicated optimization landscapes efficiently, and a learning rate scheduler (ReduceLROnPlateau) dynamically adjusts the learning rate as a function of observed patterns in test loss for stable and efficient convergence. The training process entails repeated processing of mini batches, movement of image data to the processing hardware Graphics Processing Unit (GPU), passing through the ResNet50 CBAM model, and computing the classification loss. Subsequently, gradient descent is utilized to update model parameters, enabling continuous improvement over N epochs (E). After completing the training process, the model proceeds to the evaluation stage, where it is tested on unseen data to analyze its ability for generalization. The model performance is assessed using a variety of key metrics, including accuracy, AUC-ROC, precision, recall, and F1-score, for a comprehensive assessment of its classification performance.

To improve discriminative feature learning, we combined CBAM with a ResNet50 backbone. The entire training pipeline of the ResNet50-CBAM model is presented in Algorithm 3.

Algorithm 3 summarizes the training process of the ResNet50-CBAM model. It involves loading pre trained ResNet50 weights, adding CBAM, initializing the classifier, and AdamW optimization with learning rate schedules. The model is evaluated on unseen samples at each epoch during training with classification metrics and statistical measures to assess convergence and generalization.

The core of the model is ResNet-50 with CBAM as shown in Fig. 3, which is a deep CNN for feature extraction in robust tumor classification. It uses residual connections to avoid vanishing gradients and enhance feature propagation. Given an input image I , the feature maps F_{res} are calculated as:

$$F_{\text{res}} = \text{ResNet50}(I) \quad (10)$$

Algorithm 3 ResNet50 + CBAM Model Training for Bone Tumor Classification

Require: Image dataset D with labels $\mathcal{Y}$, batch size B , learning rate η , weight decay λ , label smoothing γ , number of epochs E

Require: Number of classes N_c , pre trained ResNet50 backbone, CBAM attention module

Model Initialization:

2: Load ResNet50 with pre trained weights, remove final fully connected layer

Add CBAM attention module to enhance feature maps

4: Define classifier: Fully connected layers with BatchNorm, ReLU, Dropout, and Softmax

Initialize model $\mathcal{M}$ with ResNet50 + CBAM

6: **Define Loss Function and Optimizer:**

Set loss function as categorical cross entropy with label smoothing γ

8: Use AdamW optimizer with learning rate η and weight decay λ

Define learning rate scheduler: ReduceLROnPlateau

10: **Model Training:**

for $epoch = 1$ to E **do**

12: Set model $\mathcal{M}$ to training mode

Initialize training loss $\mathcal{L}_{\text{train}} = 0$, correct predictions $C_{\text{train}} = 0$, total samples $N_{\text{train}} = 0$

14: Start training timer

for each batch (X_b, Y_b) in D_{train} **do**

16: Move (X_b, Y_b) to device

Zero gradients of optimizer

18: Compute model output: $\hat{Y}_b = \mathcal{M}(X_b)$

Compute loss: $\mathcal{L} = \text{CrossEntropyLoss}(\hat{Y}_b, Y_b)$

20: Perform backpropagation and update weights

Accumulate training loss and correct predictions

22: **end for**

Stop training timer

24: Compute training accuracy: $\text{Acc}_{\text{train}} = \frac{C_{\text{train}}}{N_{\text{train}}} \times 100$

Model Evaluation:

26: Set model $\mathcal{M}$ to evaluation mode

Initialize test loss $\mathcal{L}_{\text{test}} = 0$, correct predictions $C_{\text{test}} = 0$, total samples $N_{\text{test}} = 0$

28: Start evaluation timer

for each batch (X_b, Y_b) in D_{test} **do**

30: Move (X_b, Y_b) to device

Compute model output: $\hat{Y}_b = \mathcal{M}(X_b)$

32: Compute loss: $\mathcal{L} = \text{CrossEntropyLoss}(\hat{Y}_b, Y_b)$

Accumulate test loss and correct predictions

34: **end for**

Stop evaluation timer

36: Compute test accuracy: $\text{Acc}_{\text{test}} = \frac{C_{\text{test}}}{N_{\text{test}}} \times 100$

Adjust learning rate using scheduler based on test loss

38: Print epoch results: Training Loss, Training Accuracy, Training Time, Test Loss, Test Accuracy, Test Time

end for

40: Compute accuracy, AUC-ROC, precision, Recall, and F1-score

Other statistical Parameters

where $F_{\text{res}} \in \mathbb{R}^{H \times W \times C}$ are feature maps of height H , width W , and channels C . The features are further enhanced using CBAM for stronger spatial and channel wise attention to improve classification accuracy.

To further improve the discriminative capacity of the feature maps extracted from the encoder, the Convolutional Block Attention Module (CBAM) is incorporated into the model. CBAM employs both channel and spatial attention mechanisms to focus on essential features while reducing the influence of irrelevant information.

The refined feature maps, denoted as F_{cbam} , are computed as follows:

$$F_{\text{cbam}} = F_{\text{res}} \cdot M_c(F_{\text{res}}) \cdot M_s(F_{\text{res}}) \quad (11)$$

where:

- $M_c(F_{\text{res}})$ represents the channel attention map, which assigns greater weight to the most important feature channels. It is computed using a combination of Global Average Pooling (GAP) and Global Max Pooling (GMP), followed by a multi layer perceptron (MLP).
- $M_s(F_{\text{res}})$ represents the spatial attention map, which highlights salient spatial regions using convolutional operations and pooling across feature maps.

This attention mechanism adaptively scales feature representations, allowing the model to focus on discriminative various bone tumor images, thereby improving classification performance.

The channel attention mechanism in CBAM boosts the most informative channel features by performing GAP and GMP operations. These operations aggregate spatial information across all pixels in a feature map, capturing global contextual dependencies. Their results are forwarded to the shared Multi Layer Perceptron (MLP), which learns the importance channel wise. The results of the two pooling operations are concatenated and forwarded to a sigmoid activation function to generate the final channel attention map.

$$M_c(F_{\text{res}}) = \sigma(\text{MLP}(\text{GAP}(F_{\text{res}})) + \text{MLP}(\text{MaxPool}(F_{\text{res}}))) \quad (12)$$

where σ is the sigmoid activation function, scaling attention values between 0 and 1. The resulting attention map is then normalized against the original feature maps F_{res} , reweighting the contribution of each channel to highlight discriminative features in tumor classification.

The spatial attention mechanism improves the detection of significant feature locations within a space. It applies average pooling and max pooling functions along the channel dimension, thus extracting both dominant and subtle features. The resulting pooled feature maps are concatenated and then fed into a convolutional layer with a 7×7 kernel size to enable the network to learn spatial significance properly. The final spatial attention map is achieved by using a sigmoid activation function:

$$M_s(F_{\text{res}}) = \sigma(\text{Conv}([\text{AvgPool}(F_{\text{res}}); \text{MaxPool}(F_{\text{res}})])) \quad (13)$$

where Conv denotes the convolutional operation, and $[\cdot]$ denotes concatenation. The spatial attention mechanism allows the model to attend to informative auricular areas, thereby suppressing background noise and enhancing classification robustness. Through the sequential use of channel and spatial attention, CBAM effectively enhances feature representation, resulting in improved classification performance in tumor detections.

After attention refinement, the feature maps go through GAP to reduce their spatial dimensions while preserving valuable information. This process results in a global descriptor vector F_{gap} , which represents the most prominent features extracted from the feature maps enhanced by CBAM:

$$F_{\text{gap}} = \text{GAP}(F_{\text{cbam}}) \quad (14)$$

where $F_{\text{gap}} \in \mathbb{R}^C$ is a compact representation with C feature channels, making it computationally efficient while maintaining strong feature aggregation. GAP effectively reduces overfitting by eliminating spatial redundancy without losing essential global information.

For classification, F_{gap} is passed through a Fully Connected (FC) layer with trainable parameters, including weight W and bias b , followed by a softmax activation function to obtain class probabilities:

$$\hat{y} = \text{softmax}(W F_{\text{gap}} + b) \quad (15)$$

where $\hat{y}$ represents the predicted class probabilities. The softmax function ensures that the outputs sum to 1, enabling probabilistic interpretation for multi class classification. This approach enhances the tumor classification model, making it both accurate and efficient.

The model is trained using the categorical cross entropy loss function, which measures the difference between the true labels y_i and the predicted probabilities $\hat{y}_i$:

$$L = - \sum_{i=1}^5 y_i \log(\hat{y}_i) \quad (16)$$

where L loss function ensures that the model learns to minimize classification errors while improving class probability predictions. The Adam optimizer is employed for model optimization with an initial learning rate of 0.0001, which balances fast convergence and stability. Training is carried out with a batch size of 32, which is memory efficient and gradient update stable. Early stopping is applied to test loss to avoid overfitting and improve generalization. Backpropagation computes the gradients of the loss function with respect to the model parameters during training. These gradients are used to update the weights W and biases b based on the Adam optimization rule, improving feature extraction and classification performance. The process is repeated until convergence or stopping conditions are met, ensuring optimal model performance.

4. Results and discussion

Each model is trained using 5421 data samples for training, while 1356 samples are utilized during the testing purpose. The performance of the suggested scheme and benchmark methodologies is assessed through a detailed explanation of the confusion matrix for each strategy. Several metrics are employed to evaluate the model's effectiveness in tumor classification. Accuracy, precision, recall, and F1-score are used in a balanced measure of false positives and false negatives. The AUC-ROC measure assesses the capacity of the model to distinguish between classes, thus providing insight into its discriminatory effectiveness. The Matthews Correlation Coefficient (MCC) and Cohen's Kappa Score are also employed to assess the reliability of the classification, particularly in imbalanced datasets. The metrics combined comprehensively evaluate the model's performance in various deformity categories.

4.1. Convolutional Neural Networks Transformer (CNNT) model

The CNNT model is highly efficient in classification, as shown in the confusion matrix in Fig. 4. The model has a virtually perfect classification of Giant Cell Tumor 99.33%, Osteofibroma 99.36%, and Synovial Osteochondroma 100%, reflecting an extremely high level of accuracy in detecting these specific forms of tumor with minimal or no error. Even with this remarkable performance, some tumor categories have misclassification because of similarities in features. For example, Osteochondroma has been misclassified into Multiple Osteochondromas 3.21% and Osteosarcoma 5.13%, which implies similar structural or morphological characteristics of these classes. Similarly, Multiple Osteochondromas have an 8.70% misclassification with Osteochondroma, reflecting the difficulty in differentiating these similar categories. One such problem is reported in Osteosarcoma, wherein 9.63% of the cases are misclassified as Osteochondroma. Likewise, Simple Bone Cyst is reported to have 7.79% misclassification as Osteochondroma, indicating the potential for overlap of features requiring tuning. These misclassifications indicate that feature extraction and representation mechanisms would be further optimized to enhance inter class separability.

The classification report in Table 1 of the CNNT model indicates exemplary performance in bone tumor classes with high precision, recall, F1-score, and specificity. Synovial Osteochondroma is ideally classified, with all the metrics at 1 and no false negatives or positives. Osteofibroma is also well classified with precision, recall, and F1-score

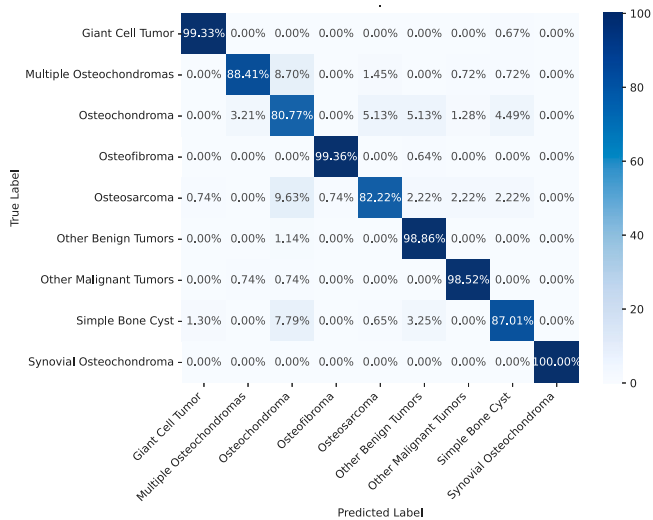


Fig. 4. Confusion matrix of CNN model.

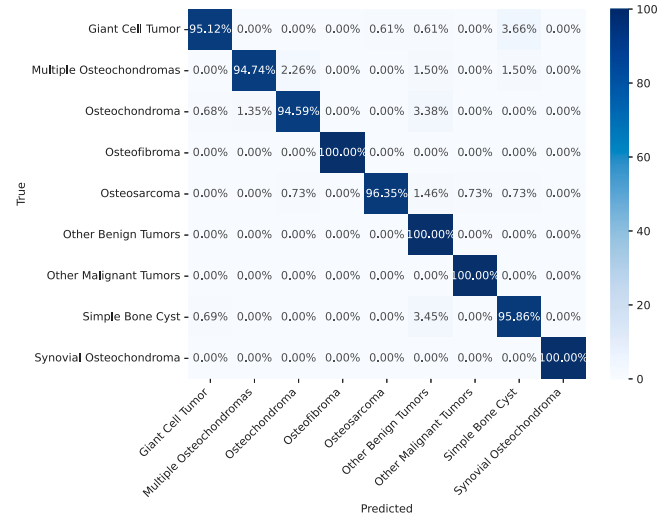


Fig. 5. Confusion matrix of Resnet50 CBAM model.

Table 1

Classification report of CNN model.

Class	Precision	Recall	F1-Score	Specificity
Giant cell tumor	0.9801	0.9933	0.9867	0.9975
Multiple osteochondromas	0.9531	0.8841	0.9173	0.9951
Osteochondroma	0.7590	0.8077	0.7826	0.9667
Osteofibroma	0.9936	0.9936	0.9936	0.9992
Osteosarcoma	0.9098	0.8222	0.8638	0.9910
Other benign tumors	0.9110	0.9886	0.9482	0.9856
Other malignant tumors	0.9568	0.9852	0.9708	0.9951
Simple bone cyst	0.9178	0.8701	0.8933	0.9900
Synovial osteochondroma	1.0000	1.0000	1.0000	1.0000

at 0.9936 and close to perfect specificity at 0.9992, which reflects precise identification.

Giant Cell Tumor has a perfect classification accuracy precision of 0.9801, recall of 0.9933, and F1-score of 0.9867, a sign of good reliability. Its 0.9975 specificity is a demonstration of its ability to reduce misclassification. Osteochondroma is challenging, as evidenced by the low F1-score (0.7826) from precision (0.7590) and recall (0.8077). A specificity of 0.9667 means that although the model correctly classifies non Osteochondroma cases, there is misclassification, particularly with Multiple Osteochondromas and Osteosarcoma, because of shared characteristics. Osteosarcoma is classified incorrectly with an F1-score of 0.8638, precision of 0.9098, and recall of 0.8222, indicating misclassification with other tumors of the same kind. Other tumor categories, including Other Benign Tumors F1-score is 0.9482, Other Malignant Tumors F1-score is 0.9708, and Simple Bone Cyst F1-score is 0.8933, maintain strong classification results, with precision and recall consistently exceeding 0.87, reinforcing the model's robustness across diverse tumor types.

4.2. ResNet50 and Convolutional Block Attention Module (CBAM) model

The ResNet50-CBAM model confusion matrix in Fig. 5 indicates excellent classification capability across different bone tumor classes with highly accurate favorable rates for most classes. The model can effectively classify tumor types with over 94% accuracy in most cases, with minimal misclassification. Osteofibroma, Other Benign Tumors, Other Malignant Tumors, and Synovial Osteochondroma are accurately labeled with 100% accuracy, i.e., the model identifies all the instances of these classes with no false negatives or false positives. For Giant Cell Tumors, the model is 95.12% accurate with very low misclassification 0.61% with Osteosarcoma and Other Benign Tumors and slightly higher, at 3.66% misclassification with Simple Bone Cyst.

However, Multiple Osteochondromas are 94.74% accurate, 2.26% misclassified as Osteochondroma and 1.50% as Other Benign Tumors or Simple Bone Cyst. These small percentages of misclassification can be accounted for by these tumors' similarity in feature representation. Osteochondroma is accurately classified with 94.59%, with very little misclassification 1.35% as Multiple Osteochondromas and 3.38% as Other Benign Tumors, showing that the model can be improved by further tuning to be able to differentiate these classes better. Osteosarcoma is correctly labeled at 96.35% and incorrectly labeled as Osteochondroma and Other Malignant Tumors at 0.73%, showing minimal overlap in the representations learned. Simple Bone Cyst is diagnosed with 95.86% accuracy, with 3.45% misdiagnosis as Other Benign Tumors, which similar morphological features can explain.

The ResNet50-CBAM model classification report in Table 2 shows better performance in bone tumor classification with high precision, recall, F1-score, and specificity. High reliability, especially for tumors with well defined types, is indicated by the results with minimal misclassifications.

Osteofibroma and Synovial Osteochondroma are classified with complete precision by the model, with all the scores of 1, which means there are no errors. Other Malignant Tumors also show nearly perfect performance: precision is 0.9936, recall is 1, and F1-score is 0.9968, showing the model's good performance in detecting malignant cases. The Osteosarcoma model has an F1-score of 0.9778 with precision at 0.9925 and recall at 0.9635, which means highly accurate classification with minimal misclassification. A 0.9992 specificity is also evidence of its strength for classifying non Osteosarcomas accurately.

Giant Cell Tumor has an F1-score of 0.9689, precision of 0.9873, and recall of 0.9512, which indicates scope for recall improvement for more accurate identification of actual cases. Multiple Osteochondromas have an F1-score of 0.9655, precision of 0.9844, and recall of 0.9474, which indicates high accuracy but minor recall problems. Osteochondroma is classified with an F1-score of 0.9589, precision of 0.9722, and recall of 0.9459. However, a high specificity of 0.9967 means the model can separate non Osteochondroma cases nicely. Other Benign Tumors also have a perfect recall 1 but less precision 0.9138 and thus an F1-score of 0.9550. Simple Bone Cyst has an F1-score of 0.9488, precision of 0.9392, and recall of 0.9586, indicating good classification but a slight false positive bias. A specificity of 0.9926 means the model can pick out the non simple Bone Cyst cases.

4.3. Comparative analysis

To evaluate the impact of the implemented preprocessing techniques, we compare the performance of the proposed models before

Table 2
Classification report of Resnet50 CBAM model.

Class	Precision	Recall	F1-Score	Specificity
Giant cell tumor	0.9873	0.9512	0.9689	0.9983
Multiple osteochondromas	0.9844	0.9474	0.9655	0.9984
Osteochondroma	0.9722	0.9459	0.9589	0.9967
Osteofibroma	1.0000	1.0000	1.0000	1.0000
Osteosarcoma	0.9925	0.9635	0.9778	0.9992
Other benign tumors	0.9138	1.0000	0.9550	0.9875
Other malignant tumors	0.9936	1.0000	0.9968	0.9992
Simple bone cyst	0.9392	0.9586	0.9488	0.9926
Synovial osteochondroma	1.0000	1.0000	1.0000	1.0000

and after applying data denoising, class balancing, and data augmentation, as shown in Table 3. The results demonstrate a substantial improvement in model precision and accuracy following these steps, thereby underscoring the critical role of preprocessing in enhancing model training and overall predictive performance.

The combination of Convolutional Block Attention Module (CBAM) with ResNet50 improved the discrimination power of the model by a large margin, taking the overall accuracy to 97.41% from 83.44%. Performance metrics of the ResNet50 baseline model in Table 4 exhibit striking variation across tumor types. CBAM was able to highlight dominant spatial and channel specific features, resulting in radical improvements in precision, recall, and F1 scores for all types of tumors. Categories previously exhibiting moderate performance, such as Osteochondroma and Osteosarcoma, exhibited considerable improvements in recall and F1-score. These findings validate the use of attention mechanisms, such as CBAM, in drastically enhancing feature localization and inter class separation, especially in intricate biomedical classification problems, thus justifying its inclusion in the proposed framework.

The ResNet50-CBAM model shows greater classification accuracy in all tumor types compared to the CNN Transformer model in Fig. 6. Although both methods are almost perfect for Osteofibroma and Synovial Osteochondroma, the ResNet50-CBAM model is more accurate in more complicated categories like Giant Cell Tumor, Osteochondroma, and Multiple Osteochondromas, efficiently reducing misclassification rates. The highest improvement in accuracy is observed in Other Benign Tumors and Simple Bone Cyst, which shows the high feature extraction and attention of ResNet50-CBAM.

The AUC-ROC values indicate that ResNet50-CBAM is better than the CNNT model for the majority of tumor types consistently, indicating greater discriminatory power in Table 5. While CNNT scores higher for Giant Cell Tumor (0.995 vs. 0.975), ResNet50-CBAM is better in the classification of Osteochondroma (0.971 vs. 0.887), Osteosarcoma (0.981 vs. 0.907), and Simple Bone Cyst (0.976 vs. 0.930), indicating its greater robustness in distinguishing complicated cases. Both models indicate perfect classification (AUC-ROC = 1.000) for Osteofibroma, Other Malignant Tumors, and Synovial Osteochondroma, proving their high reliability in these classes.

Cohen's Kappa statistic (κ) was introduced by J. A. Cohen [30] in psychology to measure the concordance between two judges. It has been used widely as a performance measure in classification issues [31] throughout the years. Cohen's Kappa (κ) estimates the concordance between predicted and actual labels, correcting for agreement by chance. This is a more significant measurement than accuracy measures. Cohen's Kappa formula is

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (17)$$

where P_o represents the observed agreement and P_e denotes the expected agreement by chance. This metric has been extensively used to evaluate classifier performance in various studies.

The MCC [32] is another robust measure for validating classification models, particularly for imbalanced datasets. Unlike accuracy, MCC considers true positives and true negatives, as well as false positives and

false negatives. The MCC formula to use in the calculation is expressed as:

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (18)$$

The statistical comparison highlights the improved effectiveness of ResNet50-CBAM over the CNNT model in terms of agreement and reliability of classification in Table 6. The Cohen's Kappa coefficient (0.9718 vs. 0.9203) and MCC (0.9719 vs. 0.9204) indicate that ResNet50-CBAM offers better interrater reliability and classification stability. Furthermore, ResNet50-CBAM substantially reduces training time (2993.63 s vs. 5692.64 s), thus improving computational efficiency and keeping testing times equal. These results support the ability of ResNet50-CBAM to offer efficient and accurate classification performance.

It is worth noting that the comparison performed in Table 7 is conducted in a binary classification setting, specifically to differentiate between Benign Tumors and Malignant Tumors. This scenario aligns with the models employed by other existing research studies, such as employing CNN architectures like ResNet50, Inception v3, and EfficientNet variants [10], as well as YOLOv8s-cls [1] and hybrid models like CNN+Standard Deviation of NN intervals (SDNN) [33] and DS-Net [12]. Under this binary scenario, our proposed CNNT model and ResNet50 combined with CBAM (ResNet50+CBAM) clearly surpass the others. The ResNet50+CBAM model achieves an Accuracy of 100.00%, a Recall of 1.0000, and an F1-Score of 0.9759, with commendable Precision (0.9537) and Specificity (0.9933), thereby leaving the next best rivals far behind. Likewise, the CNNT model demonstrated an Accuracy of 98.54%, which indicates the strength and steadiness of our methods compared to conventional methods such as CNN (ResNet50), CNN EfficientNet version B3 (EfficientNet-b3), YOLOv8s-cls [1], DS-Net [12], and others.

While the results shown in Table 7 are focused on binary classification as a benchmark, note that our built models have been applied and tested in a more general multi class classification problem, which includes nine various bone tumor types. The overall results and evaluation metrics relevant in the multi class setting are shown elsewhere in the manuscript in the form of particular tables and figures.

4.4. Qualitative analysis using grad-CAM

To acquire a better insight into the selected decision making process by deep learning models, we incorporated Gradient weighted Class Activation Mapping (Grad-CAM) visualizations, as illustrated in Fig. 7. Fig a is the original radiographic images sample for each bone tumor class. Fig b is the corresponding Grad CAM heatmaps produced by the CNNT model. In contrast, Fig c is the Grad CAM outcomes derived from the ResNet50-CBAM model. Grad CAM is an effective interpretive technique that annotates input image spatial locations with the most significant impact on a model's prediction outputs. By computing the gradient of the target class score concerning the final convolutional layer, Grad CAM generates class specific localization maps that visually depict where the model is focusing while making predictions. Warmer areas (e.g., red and yellow) in these visualizations indicate greater association with the classification task, and cooler areas (i.e., blue) are less associated.

The performance of this visualization is that ResNet50 augmented by CBAM always directs attention more precisely and intensely towards anatomical regions of interest for tumors than the CNNT model, which occasionally directs attention towards irrelevant extraneous background structures. This directed attention exhibited by the augmented model is due to its channel and spatial attention mechanisms, which systematically adjust feature responses and enhance interpretability [42]. Such augmented capability is consistent with the findings of earlier research, where architectures with embedded attention mechanisms enhanced both performance and clinical transparency in medical imaging tasks [43].

Table 3

Performance comparison of ResNet50-CBAM and CNNT models before and after preprocessing.

Class	ResNet50-CBAM (Before)				ResNet50-CBAM (After)				CNNT (Before)				CNNT (After)			
	Prec.	Rec.	F1	Spec.	Prec.	Rec.	F1	Spec.	Prec.	Rec.	F1	Spec.	Prec.	Rec.	F1	Spec.
Giant cell tumor	0.6812	0.6523	0.6623	0.8111	0.9873	0.9512	0.9689	0.9983	0.6328	0.6145	0.6212	0.7915	0.9801	0.9933	0.9867	0.9975
Multiple osteochondromas	0.8216	0.7812	0.8018	0.8422	0.9844	0.9474	0.9655	0.9984	0.8025	0.7625	0.7831	0.8311	0.9531	0.8841	0.9173	0.9951
Osteochondroma	0.8311	0.8532	0.8411	0.7932	0.9722	0.9459	0.9589	0.9967	0.8045	0.8423	0.8265	0.7712	0.7590	0.8077	0.7826	0.9667
Osteofibroma	0.7212	0.7015	0.7134	0.8534	1.0000	1.0000	1.0000	1.0000	0.6825	0.6612	0.6727	0.8022	0.9936	0.9936	0.9936	0.9992
Osteosarcoma	0.8015	0.8217	0.8116	0.8317	0.9925	0.9635	0.9778	0.9992	0.7618	0.7228	0.7427	0.8122	0.9098	0.8222	0.8638	0.9910
Other benign tumors	0.6525	0.6310	0.6422	0.8511	0.9138	1.0000	0.9550	0.9875	0.6012	0.6111	0.6012	0.7119	0.9110	0.9886	0.9482	0.9856
Other malignant tumors	0.7013	0.6815	0.6916	0.8616	0.9936	1.0000	0.9968	0.9992	0.6517	0.6317	0.6418	0.8118	0.9568	0.9852	0.9708	0.9951
Simple bone cyst	0.7617	0.7917	0.7718	0.8017	0.9392	0.9586	0.9488	0.9926	0.7214	0.7522	0.7312	0.7712	0.9178	0.8701	0.8933	0.9900
Synovial osteochondroma	0.8512	0.8215	0.8315	0.8515	1.0000	1.0000	1.0000	1.0000	0.7812	0.7614	0.7715	0.8015	1.0000	1.0000	1.0000	1.0000

Table 4

Comparative classification performance between baseline ResNet50 and ResNet50 with CBAM across different tumor classes.

Class	Baseline ResNet50				ResNet50 + CBAM			
	Precision	Recall	F1-Score	Specificity	Precision	Recall	F1-Score	Specificity
Giant cell tumor	0.8155	0.9257	0.8671	0.9743	0.9873	0.9512	0.9689	0.9983
Multiple osteochondromas	0.8000	0.7582	0.7785	0.9759	0.9844	0.9474	0.9655	0.9984
Osteochondroma	0.5745	0.7826	0.6626	0.9343	0.9722	0.9459	0.9589	0.9967
Osteofibroma	0.9321	0.9742	0.9527	0.9908	1.0000	1.0000	1.0000	1.0000
Osteosarcoma	0.9505	0.5749	0.7164	0.9958	0.9925	0.9635	0.9778	0.9992
Other benign tumors	0.9115	0.6478	0.7574	0.9916	0.9138	1.0000	0.9550	0.9875
Other malignant tumors	0.9635	0.9429	0.9531	0.9959	0.9936	1.0000	0.9968	0.9992
Simple bone cyst	0.7021	0.9362	0.8024	0.9539	0.9392	0.9586	0.9488	0.9926
Synovial osteochondroma	0.9870	0.9806	0.9838	0.9983	1.0000	1.0000	1.0000	1.0000

Table 5

AUC-ROC scores per class across different models.

Class	ResNet50-CBAM	CNNT
Giant cell tumor	0.975	0.995
Multiple osteochondromas	0.973	0.940
Osteochondroma	0.971	0.887
Osteofibroma	1.000	0.996
Osteosarcoma	0.981	0.907
Other benign tumors	0.994	0.987
Other malignant tumors	1.000	0.990
Simple bone cyst	0.976	0.930
Synovial osteochondroma	1.000	1.000

Table 6

Comparison of Cohen's Kappa score and MCC across different models.

Model	Cohen's Kappa score	MCC score	Training time (s)	Testing time (s)
ResNet50-CBAM	0.9718	0.9719	2993.63	432.40
CNNT	0.9203	0.9204	5692.64	355.58

4.5. Sensitivity analysis on dataset distribution shifts

To determine the effect of variations in data distribution, we did center wise analyses replicating real world variation between different data centers (Center 1, Center 2, and Center 3) on model generalization. Center to center test loss indicates the diversity of datasets. ResNet50-CBAM was more generalizable and stable across centers due to its attention mechanisms effectively accommodating inter center variation as shown in Table 8.

This comparison was necessitated by the intrinsic heterogeneity of data between the different centers, Center 1 possessing a considerably broader and more evenly spread range of tumors than Centers 2 and 3. For instance, rare tumor classifications like Osteofibroma and Synovial Osteochondroma are not represented or are underrepresented in Centers 2 and 3. Such a bias causes distributional imbalances that would adversely impact model performance, particularly when models trained on a center's data are applied to a different center's data.

Our experimental results further underscore the significance of data heterogeneity across centers. In particular, both CNNT and ResNet50-CBAM models demonstrated decreased performance when trained on data from one center alone and tested on data from other centers this was especially drastic when models trained on data from Center 1 were tested on Centers 2 and 3. The observed decrease in performance can be explained by distributional shifts and the weak representation of certain tumor classes in Centers 2 and 3. On the other hand, models trained and tested within Center 1 obtained relatively better accuracy and F1-scores, likely due to the larger sample size and better distribution of tumor types in that center.

To enhance the model's generalization, we could further train on the union of data from all three centers (Centers 1+2+3). When separately tested on each center, this led to substantial improvement in the key metrics such as accuracy, F1-score, and specificity. ResNet50-CBAM always performed better than CNNT in all cross center tests, being more stable against data distribution variations and more capable of generalization across different clinical data sources than models we experimented with. These results are consistent with the results of previous research. For instance [44], strongly recommended evaluating deep learning models under data heterogeneity in medical imaging, especially if data are from multiple institutions. However [45] also demonstrated how using samples from diverse domains during training greatly improves model generalization and stability under domain shift for medical diagnosis tasks.

5. Conclusion and future work

This paper presents an indepth deep learning model addressing the classification of nine significant subtypes of bone tumors from radiographic images. Based on applying a hybrid architecture of CNN and Transformer models (CNNT) and an improved ResNet50 architecture with a Convolutional Block Attention Module (CBAM), the proposed models depict considerable improvements in feature representation accompanied by classification accuracy. Applying BM3D denoising methods and data balancing with SMOTE's assistance further contributes to the quality and resilience of the training. Empirical assessments indicate that our models outperform existing state of the art models such as YOLOv8s-cls, EfficientNet, and DS-Net on all major

Table 7

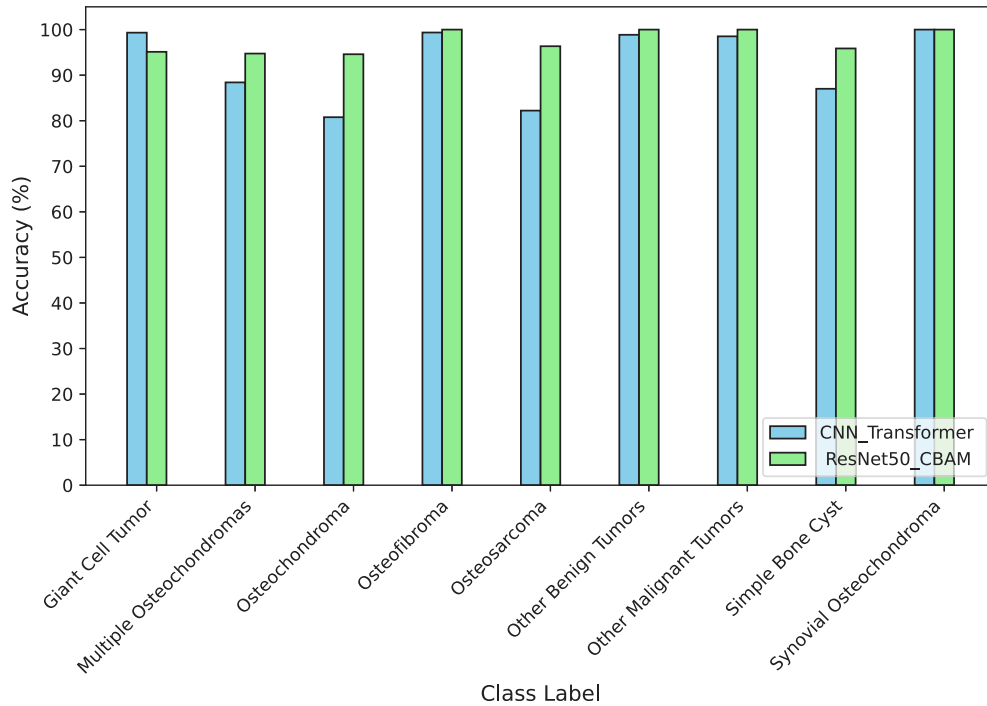
Performance comparison with existing models.

Method	Accuracy	Precision	Recall/Sensitivity	F1-Score	Specificity
CNNT model (Proposed)	98.54%	0.9339	0.9869	0.9595	0.9595
ResNet50 + CBAM model (Proposed)	100.00%	0.9537	1.0000	0.9759	0.9933
CNN (ResNet50) [10]	81.0% $\pm$ 4.6	0.76–0.80	0.74–0.78	0.75–0.80	0.86–0.89
CNN (Inception v3) [10]	82.1% $\pm$ 2.9	0.77–0.81	0.76–0.79	0.76–0.78	0.87–0.89
CNN (EfficientNet-b1) [10]	83.5% $\pm$ 5.4	0.79–0.83	0.77–0.81	0.78–0.80	0.88–0.90
CNN (EfficientNet-b2) [10]	85.3% $\pm$ 5.0	0.82–0.85	0.80–0.84	0.81–0.83	0.89–0.92
CNN (EfficientNet-b3) [10]	84.2% $\pm$ 4.0	0.81–0.84	0.78–0.81	0.79–0.81	0.88–0.90
YOLOv8s-cls [1]	~93.2%	~0.80–0.83	~0.85–0.88	~0.82–0.85	~0.89–0.91
CNN [34]	~73.4%	~0.68–0.72	~0.70–0.74	~0.70–0.73	~0.80–0.83
Multi seg-UNet [11]	~97.8%	~0.93–0.95	~0.96–0.98	~0.94–0.96	~0.98–0.99
VGG-16 [7]	~0.82–0.83	~0.77–0.80	0.8688	0.908	~0.79–0.82
DNN [35]	~81.0%	~0.75–0.78	~0.77–0.80	~0.76–0.79	~0.84–0.87
EfficientNet [11]	~76.0%	~0.70–0.73	0.79	~0.73–0.75	0.75
CNN [36]	97.38%	0.969	0.974	0.970	0.968
CNN + SDNN [33]	~95.0%	~0.91–0.94	0.9317	~0.92–0.93	0.961
CLSTM [37]	~81.0%	~0.76–0.79	~0.74–0.78	~0.75–0.78	~0.85–0.88
CNN [38]	92.20%	~0.88–0.90	~0.91–0.93	0.923	~0.91–0.93
DS-Net [12]	95.10%	0.929	0.920	0.922	0.961
CNN [39]	92.40%	0.970	0.940	0.950	~0.91–0.93
CNN [12]	91.20%	~0.89–0.91	~0.87–0.90	~0.88–0.90	~0.90–0.92
ANN [40]	80.00%	~0.70–0.73	0.75	~0.72–0.74	~0.82–0.85
EfficientNetB7 [41]	62.77%	0.2554	0.2554	~0.25–0.28	~0.55–0.60
InceptionResNetV2 [41]	93.29%	0.8658	0.8658	~0.86–0.87	~0.88–0.90
NasNetLarge [41]	90.91%	0.8182	0.8182	~0.81–0.82	~0.87–0.89
CNN (ResNet50) [41]	89.83%	0.7991	0.7992	~0.79–0.80	~0.85–0.88

Table 8

Sensitivity analysis under center wise dataset splits showing model performance and inference efficiency.

Model	Train center	Test center	Accuracy (%)	F1-score	Precision	Recall	Specificity
Resnet50 CBAM	Center all	Center all	97.41	0.9857	0.9760	0.9741	0.9969
Resnet50 CBAM	Center 1	Center 1	94.80	0.9480	0.9501	0.9472	0.9893
Resnet50 CBAM	Center 1	Center 2	84.60	0.8234	0.8120	0.8155	0.8741
Resnet50 CBAM	Center 1	Center 3	89.75	0.8720	0.8652	0.8611	0.9185
CNNT	Center All	Center All	92.56	0.9285	0.9313	0.9249	0.9911
CNNT	Center 1	Center 1	89.15	0.8923	0.8872	0.8851	0.9764
CNNT	Center 1	Center 2	76.80	0.7345	0.7120	0.7060	0.8105
CNNT	Center 1	Center 3	84.30	0.8134	0.8021	0.8012	0.8592

**Fig. 6.** Model accuracy across different classes.

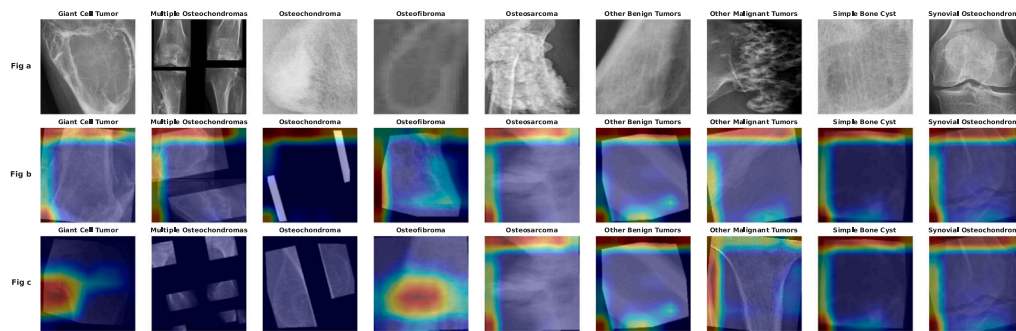


Fig. 7. Fig a Original radiographic images; Fig b presents the Grad CAM heatmaps generated by the CNNT model; Fig c displays Grad CAM outputs from the ResNet50 CBAM model. The heatmaps highlight class discriminative spatial regions that the models focus on during inference or classification. Red and yellow regions indicate areas of higher importance in the model's prediction.

performance indicators, achieving up to 100% classification accuracy in binary configurations and an F1-score value of 0.9759 in the multi class configuration. The results verify the clinical applicability of the proposed approach toward aiding radiologists in accurate and timely musculoskeletal tumor diagnosis.

Notably, the system design optimizes generalizability, efficiency, and interpretability, thus being maximally deployable in real world clinical workflows. With an AI powered decision support tool, this effort holds immense potential for reducing diagnostic burden and improving patient outcomes in oncology directed care settings. In forthcoming activities, the incorporation of federated learning will likely enable privacy preserving training of models among various healthcare centers and hence solve the problem of data security issues commonly hindering the uptake of medical AI. In addition, we aim to research neuromorphic computing using Spiking Neural Networks (SNNs) to accelerate the energy consumption and latency of real time classification models and allow integration into edge and mobile diagnostic applications. These breakthroughs are set to serve as the impetus towards the deployability of AI solutions on a larger scale within general and pediatric health care and permit earlier interventions with better planning.

CRedit authorship contribution statement

Syed Mohsin Bokhari: Writing original draft, Visualization, Software, Data curation, Methodology Formal analysis, Investigation, Conceptualization. **Muhammad Shafi:** Writing review & editing, Validation, Supervision, Resources, Project administration, Investigation, Formal analysis.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used chatgpt in order to improve the language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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